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3rd year statistical project

Evaluation of Classical Metrics and Optimal Transport for the Comparison of Dose Maps in Radiotherapy

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Acknowledgments

Abstract

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Glossary

Term	Definition
Dosymetry	In radiation therapy, dosimetry refers to the science and technology of quantitatively measuring and calculating the energy deposited by ionizing radiation in tissues or materials during the treatement. Its core objective is to accurately assess and control the radiation dose received by the patient, in order to ensure therapeutic effectiveness while minimizing damage to normal tissues.
Dose Map	A dose map is a visual representation that shows how radiation is distributed throughout a patient's body during treatment. It allows medical teams to verify that the tumor receives the right amount of radiation while the surrounding healthy tissues are spared as much as possible.
Monte Carlo (MC) simulations	Monte Carlo simulations are a computer-based method that uses a large number of random calculations to obtain numerical results. Applied to radiation therapy, this approach simulates millions of individual particle interactions to predict how radiation behaves when passing through the human body, providing a highly accurate estimate of the dose actually received by the patient.
OT (Optimal Transport)	Optimal transport theory is a mathematical optimization framework that studies how to transform one probability distribution into another while minimizing the total transportation cost under a given cost function.
Wasserstein Distance	The Wasserstein distance is a metric between probability distributions defined within the framework of optimal transport theory. It measures the distance between two distributions by the minimal cost of moving one into the other.
SSIM (Structural Similarity)	The SSIM is a metric used to assess the perceptual similarity between two images. Unlike simple pixel-by-pixel comparisons, it takes into account luminance, contrast, and structure, making it closer to how the human eye perceives image quality.
T-test	A t-test is a statistical test used to determine whether there is a significant difference between the means of two groups. It assesses whether the observed difference is likely due to chance or reflects a true underlying effect, based on the variability within the data and the sample size.
Moments	In statistics, moments are quantitative measures that describe the shape and characteristics of a probability distribution. The most commonly used are the mean (first moment), which describes the central tendency and the variance (second moment), which measures the spread of the data.

Introduction

1 Research question and context

Cancer has been a major health problem worldwide. Radiation therapy is one of the main treatments for cancer, a method that uses radiation to destroy cancer cells by blocking their ability to multiply. In this project, we mainly focus on dosimetry, which is crucial in radiotherapy because it allows clinicians to precisely measure and control the radiation dose given to patients.

The comparison of dose maps is an essential step in dosimetry, as it ensures the accuracy of the simulated dose distributions. Each value on a dose map generally represents the average of a set of samples generated by Monte Carlo (MC) simulations [**?**]. MC simulations track millions of radiation particles as they travel through the body. Each particle can be absorbed by tissues, scattered in different directions, or pass through without any interaction. These events are determined by physical probabilities. By adding up the energy deposited in each region of the body, dosimetry creates detailed dose maps that show the average radiation dose received in different areas. This information is essential for treatment planning because it helps doctors ensure that cancerous tissues receive enough radiation to destroy tumor cells while keeping exposure to surrounding healthy organs and tissues as low as possible. Accurate dosimetry is therefore key to finding the right balance between effective treatment and patient safety, which improves survival rates and reduces the risk of radiation side effects and other complications.

2 State of the art in dosimetry

The central question addressed in this project is whether the metrics currently used to compare dose maps in radiotherapy are sufficient to capture all clinically relevant discrepancies between two Monte Carlo (MC) simulations, and whether alternative approaches based on distributional comparison can provide additional diagnostic information.

In a typical dosimetric workflow, two dose maps are compared in order to validate a new simulation algorithm, assess the impact of a change in acquisition parameters, or verify the consistency between a reference plan and a delivered treatment. Each pixel of a dose map stores the estimated mean dose derived from a set of MC particle histories. The comparison metrics are then applied to these mean dose maps, pixel by pixel or globally, to quantify the level of agreement between the two simulations.

2.1 Mean Absolute Error and Mean Squared Error

The most elementary metrics are the Mean Absolute Error (MAE) and the Mean Squared Error (MSE), defined respectively as:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i,j} \left| d^1_{i,j} - d^2_{i,j} \right| \tag{1}$$

$$\mathrm{MSE} = \frac{1}{N} \sum_{i,j} \left(d^1_{i,j} - d^2_{i,j} \right)^2 \tag{2}$$

where $d^1_{i,j}$ and $d^2_{i,j}$ denote the mean dose values at pixel (i, j) for the reference and evaluated simulations respectively, and N is the total number of pixels. These metrics are straightforward to compute and provide a global measure of the average discrepancy between two dose maps. However, they are purely pointwise: they compare mean dose values without accounting for the statistical uncertainty associated with each estimate. In particular, MSE penalises large deviations more heavily than small ones through its quadratic structure, while MAE treats all deviations equally. Neither metric provides spatial information about where the discrepancies are located, nor do they distinguish between a systematic bias and random noise.

2.2 The Gamma Index

The gamma index, introduced by Low et al. (1998), is the reference clinical metric for dose map comparison [**?**]. For each pixel of the evaluated map, it computes a combined criterion that accounts simultaneously for the local dose difference and the spatial displacement between corresponding dose values. Formally, for a reference dose distribution $D_r(\mathbf{r}_r)$ and an evaluated distribution $D_e(\mathbf{r}_e)$, the gamma value at a reference point $\mathbf{r}_r$ is defined as:

$$\gamma(\mathbf{r}_r) = \min_{\mathbf{r}_e} \left\{ \sqrt{\frac{|\mathbf{r}_e - \mathbf{r}_r|^2}{\Delta d^2} + \frac{|D_e(\mathbf{r}_e) - D_r(\mathbf{r}_r)|^2}{\Delta D^2}} \right\} \tag{3}$$

where Δd is the distance-to-agreement (DTA) tolerance and ΔD is the dose difference tolerance, typically set to 3 mm and 3% respectively in clinical practice. A pixel passes the gamma criterion if $\gamma \leq 1$. The overall pass rate, i.e. the proportion of pixels satisfying this condition, summarises the agreement between the two maps.

The gamma index is widely adopted because it incorporates clinical intuition: a small spatial shift in a region of high dose gradient may produce a large numerical dose difference but correspond to an acceptable geometric deviation. By jointly tolerating dose and distance discrepancies, the gamma index avoids penalising such situations. Its main limitation, however, is its binary nature: a pixel either passes or fails, with no distinction between a value of $\gamma = 0.1$ and $\gamma = 0.9$. Subtle but spatially structured biases that fall systematically just below the acceptance threshold may yield a high pass rate while still being clinically significant.

2.3 Structural Similarity Index

The Structural Similarity Index (SSIM), originally developed for image quality assessment [**?**], measures the perceptual similarity between two images by decomposing the comparison into three components: luminance, contrast, and structural information. For two image patches x and y of the dose map, it is defined as:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{4}$$

where μ_x, μ_y are the local means, σ_x^2, σ_y^2 the local variances, σ_{xy} the cross-covariance, and c_1, c_2 are small stabilising constants. The SSIM takes values in $[-1, 1]$, with 1 indicating perfect structural agreement.

Unlike MAE and MSE, SSIM captures spatial correlations and is sensitive to structural patterns in the dose distribution. It is therefore better suited to detecting coherent spatial discrepancies such as shifted dose gradients or altered dose fall-off regions. However, SSIM operates on the mean dose values and does not model the stochastic nature of MC simulations. It is also primarily designed for visual image comparison and does not have a direct clinical interpretation in dosimetric terms.

2.4 Student's T-test

The Student t-test provides a statistical framework for assessing whether the mean dose values across two maps differ significantly. Given two samples of pixel-level dose differences, the test evaluates the null hypothesis that the mean difference is zero against the alternative that it is non-zero. The test statistic is:

$$t = \frac{\bar{\delta}}{\hat{\sigma}_\delta / \sqrt{N}} \tag{5}$$

where $\bar{\delta}$ is the sample mean of the pixel-wise differences $d_{i,j}^1 - d_{i,j}^2$, $\hat{\sigma}_\delta$ is their sample standard deviation, and N is the number of pixels. Under the null hypothesis, this statistic follows a Student distribution with $N - 1$ degrees of freedom.

The t-test provides a principled measure of global statistical significance. It can detect whether a systematic offset exists between two dose maps even when the pointwise differences are small. Its main limitation is that it provides no spatial resolution: a single p-value summarises the entire map, making it impossible to localise the source of any detected discrepancy. Moreover, it evaluates differences in mean dose only, and is insensitive to differences in higher-order distributional characteristics such as variance or skewness.

2.5 Summary and Motivation

Table 1 summarises the key properties of the metrics reviewed above. All five metrics share a common limitation: they reduce the MC output at each pixel to a single scalar, typically the sample mean dose, and compare these scalar summaries. This design choice discards the distributional information carried by the full set of MC particle histories, in particular the local statistical uncertainty encoded in the sample variance and higher-order moments.

Metric	**Spatial resolution**	**Distributional sensitivity**	**Clinical interpretation**
MAE	pixel-wise	mean only	moderate
MSE	pixel-wise	mean only	moderate
Gamma index	pixel-wise (binary)	mean only	high
SSIM	local patch	mean + structure	low
t-test	global	mean only	moderate

Table 1: Summary of the classical metrics used in dosimetry and their main properties.

None of these metrics exploits the full empirical distribution of MC samples at each pixel. In particular, they are insensitive to differences in local sampling variability, which is a significant source of information in MC simulations where the number of particle histories per pixel varies substantially across the map. This observation motivates the investigation of metrics that compare full distributions rather than summary statistics, which is the subject of the following section.

3 Transportation Theory and Wasserstein Distance for Dosimetry

3.1 Limitations of Classical Metrics in Dosimetry

As reviewed in the previous section, the metrics most commonly used for dose map comparison (MAE, MSE, the gamma index, SSIM, and the Student t-test) share an important limitation: they operate on scalar summary values at each pixel, typically the estimated mean dose, without taking into account the underlying statistical distribution of the Monte Carlo (MC) samples from which these estimates are derived.

In a MC simulation, the dose value at each pixel is the empirical mean of a large collection of stochastic particle histories. The number of histories per pixel is not uniform across the simulation domain: pixels that are directly exposed to the radioactive ray accumulate far more particle interactions than peripheral or shielded regions, which results in strongly heterogeneous sample sizes across the map. A pixel with very few samples carries a much larger statistical uncertainty than one with thousands of samples, yet classical metrics treat both identically by assigning the same weight to their mean dose values. This insensitivity to local sampling variability is a known limitation of MAE and MSE in dosimetry.

The gamma index partially addresses spatial tolerance by introducing acceptance criteria on both dose difference and distance-to-agreement, but it remains threshold-based and binary: a pixel either passes or fails. Subtle, spatially coherent biases that fall just below the acceptance threshold may go undetected even if they are clinically meaningful. SSIM captures structural patterns but does not model the stochastic nature of MC uncertainty. The Student t-test evaluates global distributional differences but provides no spatial resolution.

In summary, none of these metrics uses the full distributional information available at each pixel. They reduce a collection of MC samples to a single number, discarding information about spread, asymmetry, and tail behaviour that may be relevant for clinical evaluation. This motivates the exploration of a different approach: one that compares the full empirical distributions of MC samples between two dose maps, rather than their pointwise summary statistics.

3.2 Optimal Transport and the Wasserstein Distance

3.2.1 The Monge-Kantorovich Problem

Optimal transport (OT) theory provides a mathematical framework for comparing probability distributions by quantifying the minimum effort required to reshape one distribution into another. The standard formulation is the Monge-Kantorovich problem: given two probability measures μ and ν defined on measurable spaces $\mathcal{X}$ and $\mathcal{Y}$, and a cost function $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^+$, the OT problem try to find a coupling measure π on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν that minimises the total transport cost:

$$W_p^p(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y)^p \, d\pi(x, y) \tag{6}$$

where $\Pi(\mu, \nu)$ denotes the set of all joint distributions on $\mathcal{X} \times \mathcal{Y}$ whose marginals are μ and ν respectively. The measure π is called a transport plan: it encodes how mass at location x in the source distribution is moved to location y in the target distribution. When $c(x, y) = |x - y|$, the p-th root of equation (6) defines the L^p-Wasserstein distance $W_p(\mu, \nu)$.

In one dimension, the Wasserstein distance has a closed-form expression using the quantile functions (inverse cumulative distribution functions) F_μ^{-1} and F_ν^{-1}:

$$W_p^p(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(t) - F_\nu^{-1}(t) \right|^p dt \tag{7}$$

This formula is the theoretical basis for the pixelwise Wasserstein computation used in this work. For each pixel, the two empirical distributions of MC samples are compared through their empirical quantile functions, which produces a scalar distance that captures both the shift in location and the differences in spread between the two distributions.

3.2.2 Geometric Interpretation

The Wasserstein distance can be understood as the minimum amount of work needed to transport the mass of one probability distribution to match another, where work is defined as mass multiplied by displacement. This interpretation (sometimes called the earth mover's distance) makes W_p sensitive to the geometry of the support of the distributions and not just to pointwise differences in probability mass.

This geometric sensitivity has two useful consequences in the dosimetric setting. First, W_p naturally accounts for the spread of the sample distribution at each pixel: a pixel with high MC variance will have a broader distribution, and the Wasserstein distance between two such distributions will reflect both any shift in mean dose and any difference in dispersion. This is precisely the information that classical metrics discard. Second, W_p is a true metric on the space of probability measures, satisfying symmetry, non-negativity, and the triangle inequality, which makes it a theoretically well-grounded comparison criterion.

3.3 Application to Dose Map Comparison

3.3.1 Pixelwise Wasserstein Mapping

In the framework developed in this project, the Wasserstein distance is computed independently at each pixel (i, j) of the dose map. Let $S_{i,j}^1 = \{x_1, \ldots, x_n\}$ and $S_{i,j}^2 = \{y_1, \ldots, y_m\}$ denote the sets of MC samples at pixel (i, j) for the reference simulation and the simulation under evaluation, respectively. The empirical distributions $\mu_{i,j}$ and $\nu_{i,j}$ are defined as the uniform discrete measures over these sample sets:

$$\mu_{i,j} = \frac{1}{n} \sum_{k=1}^{n} \delta_{x_k}, \qquad \nu_{i,j} = \frac{1}{m} \sum_{k=1}^{m} \delta_{y_k} \tag{8}$$

where δ_x denotes the Dirac mass at x. The pixelwise Wasserstein map is then defined as:

$$W_{\text{map}}(i, j) = W_1(\mu_{i,j},\, \nu_{i,j}) \tag{9}$$

This produces a two-dimensional map in which each pixel value encodes the distributional discrepancy between the two simulations at that location. Pixels where the two simulations agree will exhibit low Wasserstein values, while pixels affected by systematic bias or heterogeneous sampling will stand out with higher values.

3.3.2 Sensitivity to Sampling Heterogeneity and Bias

A key motivation for using the Wasserstein distance in this context is its sensitivity to two distinct sources of discrepancy between dose maps: additive bias and sampling variability. Classical metrics such as MAE and MSE mainly detect shifts in mean dose, while remaining insensitive to differences in the width or shape of the sample distribution. The Wasserstein distance penalises both.

In the simulated dose maps used in this study, a controlled additive bias is introduced in a specific spatial region (columns 30 to 35 of the map), while the rest of the map is generated from identical underlying distributions but with heterogeneous sample sizes. This setup mimics the spatial non-uniformity of real MC simulations. The experiments show that the Wasserstein map successfully highlights both the biased region and the zones with strongly differing sample sizes, while MAE and MSE only detect the biased zone and remain blind to sampling heterogeneity.

This sensitivity to sampling variability is particularly valuable in clinical dosimetry. In regions outside the primary radioactive ray or deep in tissue, the number of particle histories is low and the associated uncertainty is high. The Wasserstein distance provides a single criterion that integrates both dose accuracy and statistical reliability, which makes it more informative than classical metrics for evaluating MC simulations.

3.4 Computational Limitations and Motivation for Moment-Based Approximation

Despite its advantages, the direct computation of the Wasserstein distance from raw MC samples faces a fundamental scalability problem. In realistic clinical MC simulations, each pixel accumulates a number of particle histories that can range from a few hundred in low-fluence regions to several millions in high-fluence zones. Storing the complete sample set for every pixel of a two-dimensional dose map is computationally intractable. A dose grid of 512×512 pixels with 10^6 samples per pixel would require storing on the order of 2.6×10^{11} floating-point values, which far exceeds the memory capacity of standard computing systems.

In practice, MC simulation codes do not store individual particle histories. What is retained are summary statistics of the deposited dose per pixel, typically the sample mean and the sample variance, together with the number of contributing histories. This is the standard output format of production MC codes such as EGSnrc, MCNP, and Geant4. As a result, the raw sample distributions that would be needed for an exact Wasserstein computation are never available in a real dosimetric workflow.

This observation raises a central question for this project: is it possible to approximate the Wasserstein distance between two pixel distributions using only compressed representations of those distributions, specifically their statistical moments? If the answer is yes, then the distributional sensitivity of the Wasserstein metric could be used in clinical dosimetry without requiring access to the full sample sets. The following section addresses this question by drawing on recent theoretical results from Mula and Nouy (2024) [**?**].

4 Approximation of the Wasserstein Distance with Moments

4.1 Statistical Moments as Compact Distributional Descriptors

The moments of a probability measure μ on $\mathbb{R}$ are defined, for integer order $k \geq 0$, as:

$$m_k(\mu) = \int x^k \, d\mu(x) = \mathbb{E}[X^k] \tag{10}$$

The sequence of all moments $(m_k)_{k\geq 0}$ encodes the full distribution under mild regularity conditions. More precisely, if the moment sequence determines μ uniquely (a property known as moment determinacy, which holds when μ has compact support), then knowledge of all moments is equivalent to knowledge of the distribution itself.

From a practical point of view, moments provide an attractive compressed representation of a distribution: instead of storing thousands or millions of samples, one retains a finite vector of scalar statistics. The first two moments, the mean $m_1 = \mathbb{E}[X]$ and the variance $\sigma^2 = \mathbb{E}[X^2] - m_1^2$, are already stored by standard MC simulation codes. Higher-order moments encode more refined information about the shape of the distribution: skewness (order 3), kurtosis (order 4), and beyond.

The central question is whether a finite truncation of the moment sequence provides a useful approximation of the Wasserstein distance. The theoretical framework of Mula and Nouy (2024) provides a rigorous affirmative answer under well-defined conditions, which are described in the following subsection.

4.2 Theoretical Framework: the Moment-SoS Approach

4.2.1 Reformulation of Optimal Transport as a Moment Problem

Mula and Nouy (2024) establish that, when the cost function c is a polynomial and the supports $\mathcal{X}_1$ and $\mathcal{X}_2$ of the two measures are compact semi-algebraic sets (sets defined by finitely many polynomial inequalities), the OT problem is exactly equivalent to a generalised moment problem. The key observation is that for a polynomial cost $c(x, y) = \sum_\alpha c_\alpha x^\alpha$, the transport cost functional is linear in the moments of the transport plan π:

$$\int c(x, y) \, d\pi(x, y) = \sum_\alpha c_\alpha \, m_\alpha(\pi) \tag{11}$$

The marginal constraints (π must have μ and ν as marginals) also translate directly into linear constraints on the moments of π. The original infinite-dimensional optimisation problem over transport plans therefore reduces to an optimisation over the moment sequence $y = (y_\alpha)$ of the transport plan, subject to linear moment constraints and the requirement that y corresponds to a valid measure. This requirement is called the moment sequence condition, $y \in \mathrm{MS}(\mathcal{X}_1 \times \mathcal{X}_2)$.

Theorem 3.1 of Mula and Nouy formalises this equivalence: under compactness of the supports, an optimal transport plan π^* is exactly the measure whose moment sequence y^* solves the moment problem, and conversely. This equivalence is the theoretical foundation that justifies working with moments rather than with the measures themselves.

4.2.2 The Sums-of-Squares Hierarchy and Finite Truncation

The moment sequence y is infinite, so the moment problem cannot be solved directly. The computational strategy of Mula and Nouy consists in considering a hierarchy of finite truncations: for a relaxation order r, only moments up to order $2r$ are retained, and the moment sequence condition is replaced by a finite set of positive semidefinite (PSD) matrix constraints on localised moment matrices.

More precisely, if $\mathcal{X}$ is a compact semi-algebraic set defined by polynomial inequalities $g_j(x) \geq 0$, then by the Putinar Positivstellensatz, the moment sequence condition is equivalent to requiring that the localised moment matrices $M_r(g_j y)$ are positive semidefinite for all j and all orders r:

$$M_r(g_j\, y) \succeq 0, \quad \forall j, \quad \forall r \in \mathbb{N} \tag{12}$$

For a fixed truncation order r, the relaxed problem involves only finitely many moment variables and finitely many PSD constraints. This makes it a semidefinite programme (SDP) that can be solved with standard numerical solvers. This yields a hierarchy of approximations ρ_r of the true Wasserstein distance ρ, indexed by the relaxation order $r = r^*, r^* + 1, r^* + 2, \ldots$

4.2.3 Convergence Guarantee

The central theoretical result of Mula and Nouy is Theorem 6.1, which establishes the convergence of this hierarchy. The theorem states the following:

1. The sequence $(\rho_r)_{r \geq r^*}$ is non-decreasing and satisfies $\rho_r \leq \rho$ for all r.
2. $\rho_r \to \rho$ as $r \to \infty$.
3. From any sequence of solutions y^r of the relaxed problems, one can extract a subsequence whose components (y^r_α) converge to the components of a moment sequence y^* that solves the original moment problem and whose representing measure is an optimal transport plan.

The proof relies on a compactness argument: the PSD constraints on the localised moment matrices imply uniform bounds on all moment variables (Lemma 6.2 in the paper), making the feasible set of the relaxed problem compact in finite dimension. By the Banach-Alaoglu theorem, any sequence of solutions then admits a convergent subsequence, and its limit satisfies all the constraints of the original problem.

From a practical standpoint, Mula and Nouy report that even the first-order relaxation ($r = 1$) produces very accurate approximations of the Wasserstein distance in their numerical experiments: relative errors of the order of 10^{-12} are observed for the W_1 and W_2 distances between translated image distributions. This suggests that the hierarchy converges rapidly in practice, even though the computational cost grows with r.

4.3 Implications for the Dosimetric Setting

4.3.1 What the Theory Justifies

The theoretical framework of Mula and Nouy justifies, under well-defined conditions, the use of moments as a compressed representation from which the Wasserstein distance can be recovered with arbitrary precision. Applied to the dosimetric context, this means that if one can compute and store a sufficient number of moments of the MC sample distribution at each pixel, rather

than the samples themselves, one retains in principle all the information needed to compute the Wasserstein distance between two dose maps.

The conditions required by the theory are compatible with the dosimetric setting. The support of the dose distribution at each pixel is bounded, since doses are non-negative and bounded by physical energy conservation arguments. This makes it a compact semi-algebraic set. The quadratic cost $c(x, y) = (x - y)^2$ corresponding to the W_2 distance is a polynomial, so Theorem 3.1 of Mula and Nouy applies directly. For the W_1 distance with cost $c(x, y) = |x - y|$, the piecewise polynomial extension of Theorem 3.6 covers this case.

4.3.2 Practical Considerations and Limitations

While the theoretical guarantees are strong, applying the moment-SoS framework to a practical pixelwise dosimetric computation raises several challenges that are investigated in this project.

A first challenge is the choice of moment type. The raw moments $\mathbb{E}[X^k]$ of a distribution supported on $[0, 1]$ decay rapidly with order k, becoming numerically indistinguishable from zero beyond order 5 to 6 for typical dosimetric distributions. This means that a naive implementation using raw moments will lose discriminative power at high orders, regardless of how many moments are computed. Centred and standardised moments,

$$\tilde{m}_k = \mathbb{E}\left[\left(\frac{X - \mu}{\sigma}\right)^k\right] \tag{13}$$

avoid this problem by removing the influence of location and scale before computing higher-order statistics, and are therefore better suited for the comparison task.

A second consideration concerns the scope of the theory. The Wasserstein distance in the Mula and Nouy framework is computed between two global measures over the full domain. The pixelwise approach used in this project is a practical simplification where the comparison is done independently at each pixel. This does not directly correspond to the theoretical setting of the paper. However, the theoretical insight that a finite number of moments can recover the Wasserstein distance remains the justification for using moment-based features at the pixel level.

A third consideration is the role of the sample size. The number of MC samples per pixel, while not a classical moment, carries important information about the reliability of the moment estimates. A mean dose estimated from 100 samples is far less reliable than one estimated from 10 000 samples. In this project, the sample size n is therefore included as an additional feature alongside the statistical moments. It serves as a proxy for the confidence associated with the distributional estimate at each pixel, which is directly relevant to the sensitivity analysis described in the following section.

4.4 Summary

The theoretical results of Mula and Nouy establish that the Wasserstein distance between two distributions is, in principle, fully recoverable from their moment sequences, with a convergence guarantee as the number of retained moments increases. This provides the mathematical justification for the moment-based approximation strategy explored in this project.

The practical implementation works as follows. For each pixel of the dose map, a vector of statistical descriptors is computed from the available MC output. This vector includes the

sample size, the sample mean, and a set of standardised higher-order moments. The Wasserstein distance between two dose maps is then approximated by comparing these descriptor vectors across corresponding pixels, without requiring access to the raw sample distributions. The following section describes the experimental setup and the numerical results obtained with this approach.

5 Experiments and Results

This section presents the numerical experiments carried out to evaluate the different metrics described in the previous sections. All experiments use simulated two-dimensional dose maps where each pixel contains a list of MC samples drawn from a known distribution. This controlled setting makes it possible to assess the ability of each metric to detect a region of interest, since the ground truth is known at all times.

5.1 Experimental Setup and Simulated Dose Maps

The simulated dose maps are generated as follows. Two 64×64 dose maps are created, where each pixel (i, j) contains a set of MC samples drawn from a normal distribution $\mathcal{N}(\mu_{i,j}, \sigma_{i,j})$. The parameters $\mu_{i,j}$ and $\sigma_{i,j}$ are drawn once and shared between the two maps, so that both maps come from the same underlying distribution at every pixel. The only differences between the two maps are the number of samples per pixel and a controlled additive bias introduced in a specific region.

The sample counts are deliberately heterogeneous to mimic the non-uniform sampling of real MC simulations. The map is divided into four concentric zones with increasing sample counts from the border to the centre: approximately 100 samples per pixel in the outer ring, 1 000 in the next ring, 5 000 in the inner ring, and up to 10 000 in the central zone. This reflects the physical reality that high-fluence regions accumulate more particle histories than peripheral or shielded regions.

A controlled additive bias b is then added to all samples in columns 30 to 34 of the second map, after which the samples are renormalised to $[0, 1]$. Two values of b are considered: a strong bias ($b = 0.15$) that produces a visible difference between the two mean dose maps, and a weak bias ($b = 0.05$) that is not directly visible when comparing the mean dose maps. All subsequent experiments use the weak bias setting, which is the more challenging and clinically relevant scenario.

Figure 1: Dose map for a weak bias $b = 0.05$ and a strong bias $b = 0.15$.

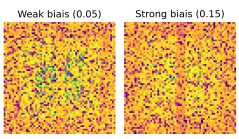

Figure **??** illustrates the effect of the bias level. The strong bias case clearly shows the biased columns as a visible vertical band, while in the weak bias case no such structure is apparent. This confirms that the bias is correctly implemented and that it represents a realistic scenario where simple visual inspection of the mean dose maps is not sufficient to detect the discrepancy.

5.2 Results of Classical Metrics

The classical metrics (MAE, MSE, SSIM, gamma index, and t-test) are applied to the weak bias setting. For each metric, the result is displayed as a two-dimensional map where each pixel value

reflects the local discrepancy between the two simulations as measured by the corresponding criterion.

Figure 3 in the appendix shows the resulting maps. The MAE and MSE maps show a slight signal in the biased columns, which can be distinguished from the rest of the map with careful inspection. However, this signal is weak and would not be conclusive in a clinical setting where the ground truth is unknown. The SSIM map, the gamma index, and the t-test do not show any structure that could be associated with the biased region: their maps are essentially uniform across the entire image.

These results illustrate the limitations discussed in Section 3. MAE and MSE have some sensitivity to the additive bias because it shifts the mean dose, but they remain blind to the sampling heterogeneity and their signal is too weak to be reliable for a small bias level. The other metrics provide no useful information in this setting. None of the classical metrics is therefore sufficient for a fine-grained evaluation of MC simulations where both weak biases and sampling variability are present.

5.3 Wasserstein Distance Computed from Raw Samples

The Wasserstein distance W_1 is computed at each pixel using the full sets of MC samples, following the pixelwise approach described in Section 3.3. The computation uses the empirical quantile formula:

$$W_1(\mu, \nu) = \int_0^1 \left| F_\mu^{-1}(t) - F_\nu^{-1}(t) \right| dt \tag{14}$$

applied to the sorted sample sets of each pixel.

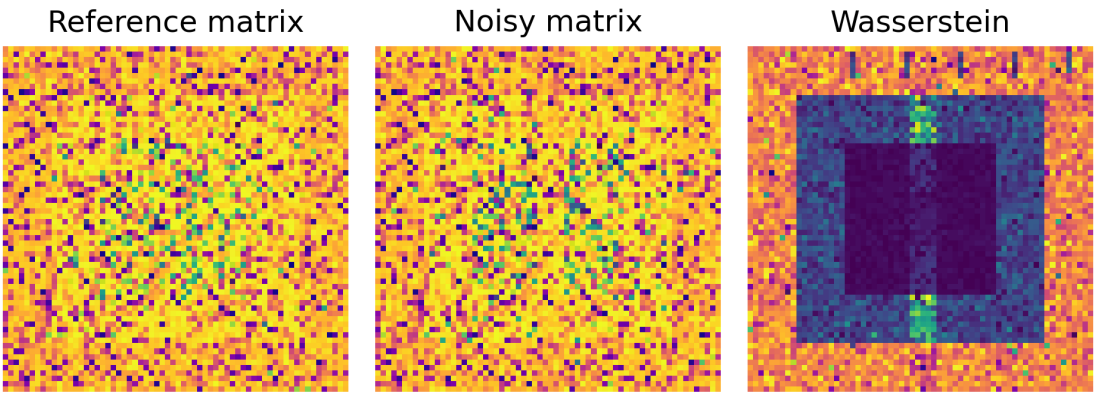

Figure 2: Pixelwise Wasserstein map computed from the full sets of MC samples.

Figure 4 shows the resulting Wasserstein map. Two distinct structures are clearly visible. First, the biased region (middle columns) stands out with elevated Wasserstein values, confirming that the metric is sensitive to the additive offset even when it is small. Second, the outer border of the map, where the sample count is lowest (around 100 samples per pixel), also shows higher values. This reflects the sensitivity of the Wasserstein distance to sampling variability: pixels with few samples produce noisier empirical distributions, and the distance between two noisy distributions is naturally higher than between two well-sampled ones.

This result demonstrates that the Wasserstein distance computed from raw samples is a powerful tool for dose map comparison. However, as discussed in Section 3.4, this approach is not feasible in practice: real MC simulation codes do not store individual particle histories, and the storage requirements would be entirely prohibitive for clinical dose grids. The map in Figure 4 therefore serves as the reference target for the approximation strategies explored in the following subsections.

5.4 Wasserstein Distance via the Moment-SoS Hierarchy

The first approximation strategy follows the theoretical framework of Mula and Nouy (2024) described in Section 4. The Wasserstein distance is approximated by solving, at each pixel, the semidefinite programme (SDP) of the moment-SoS hierarchy at relaxation order $r = 2$. This requires only the first $2r = 4$ moments of each pixel distribution, which are extracted from the pre-computed moment matrices.

Concretely, for a pixel with moment sequences $\mu = (m_0, m_1, m_2, m_3, m_4)$ and $\nu = (n_0, n_1, n_2, n_3, n_4)$,

the SDP minimises:

$$\min_{y} \; y_{(2,0)} - 2\, y_{(1,1)} + y_{(0,2)} \tag{15}$$

subject to marginal constraints $y_{(k,0)} = m_k$ and $y_{(0,k)} = n_k$ for $k = 0, \ldots, 4$, and the Putinar positive semidefinite constraints on $[0,1]^2$ as described in Theorem 3.4 of the paper. This is solved using the `cvxpy` library with the SCS solver.

Figure **??** shows the result. The biased region is clearly visible in the zones where the sample count exceeds 1,000 per pixel, which corresponds to the three inner concentric regions of the map. In the outer border, where pixels have only around 100 samples, the signal is too weak to distinguish the biased columns. The sampling heterogeneity between the concentric zones is not clearly recovered, and the map does not reproduce this aspect of the reference Wasserstein map of Figure 4.

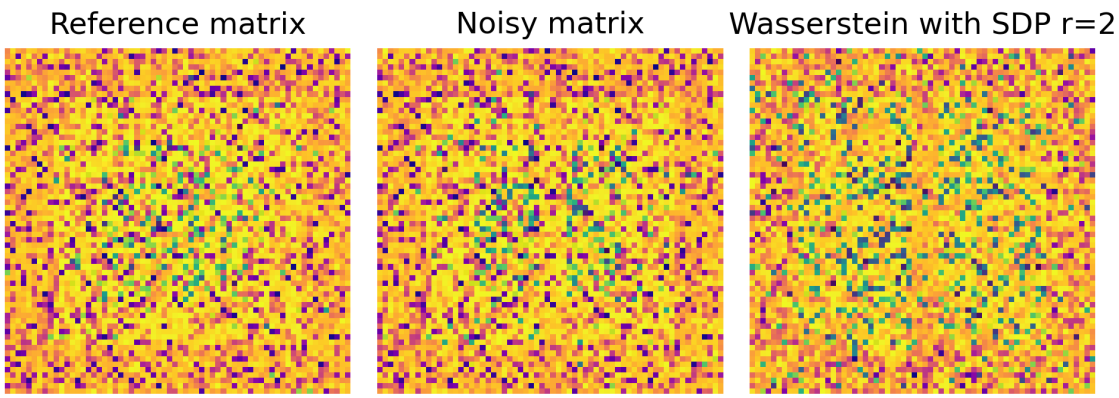

Figure 3: Pixelwise map produced by the Wasserstein distance approximated via the moment-SoS SDP at relaxation order $r = 2$.

This result partially validates the theoretical approach of Mula and Nouy in the dosimetric setting: with only 4 moments per pixel, the SDP recovers a useful signal in the well-sampled regions of the map. According to the convergence guarantee of Theorem 6.1, increasing the relaxation order r would incorporate more moments and could improve the result. However, the computational cost of the SDP already makes the $r = 2$ case slow for a 64×64 map, and higher values of r were not tested in this project for this reason. Scaling this approach to larger dose grids or to three-dimensional volumes in a clinical setting would require significant algorithmic optimisation.

5.5 MAE of Selected Moment Combinations

A lighter alternative to the SDP approach is explored here. Instead of solving an optimisation problem at each pixel, the comparison is reduced to a single scalar by averaging a selected subset of moments, and the MAE is then computed between these averaged values across the two maps. For a selected set of moment indices $\mathcal{K} = \{k_1, k_2, \ldots, k_p\}$, the representative value at pixel (i, j) is:

$$\bar{m}_{i,j}(\mathcal{K}) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} m_k^{(i,j)} \tag{16}$$

and the comparison map is the pixelwise absolute difference:

$$D(i, j) = \left| \bar{m}_{i,j}^{(1)}(\mathcal{K}) - \bar{m}_{i,j}^{(2)}(\mathcal{K}) \right| \tag{17}$$

Two choices of $\mathcal{K}$ are considered. The first is a set of evenly spaced indices covering the full range of available moments: $\mathcal{K}_1 = \{1, 20, 40, 60, 80, 100\}$. The second is a set of six randomly chosen indices always including the first moment: $\mathcal{K}_2 = \{1, 3, 4, 33, 81, 98\}$.

Figure 6 in the appendix shows the resulting maps. With $\mathcal{K}_1$, the biased region is slightly visible in the well-sampled zones, at a level comparable to what MAE achieves on the raw mean dose maps. The sampling heterogeneity is not recovered. With $\mathcal{K}_2$, the result is poor: neither the biased region nor the sampling heterogeneity can be distinguished, and the map is essentially uniform.

Why dispersed moment selections perform better?

The difference in performance between $\mathcal{K}_1$ and $\mathcal{K}_2$ can be understood by looking at the structure of the raw moment sequence. For a distribution supported on $[0, 1]$, the raw moments $m_k = \mathbb{E}[X^k]$ form a strictly decreasing sequence that decays toward zero as k increases. This property follows directly from the fact that $x^k \leq x^{k-1}$ for all $x \in [0, 1]$, so $\mathbb{E}[X^k] \leq \mathbb{E}[X^{k-1}]$. As a result, the informative content of the moment sequence is not uniformly distributed across orders: low-order moments are dominated by the mean of the distribution, while intermediate-order moments carry information about the shape of the distribution, before eventually becoming numerically negligible at high orders.

When the selected indices are concentrated at low orders, as is the case for $\mathcal{K}_2$ where three out of six indices are below 5, the averaged value $\bar{m}(\mathcal{K})$ is dominated by the first moment and essentially recovers the mean dose. This brings the method back to the same limitation as classical MAE. When the indices are spread across the full range, as in $\mathcal{K}_1$, the average captures contributions from moments at different levels of the distributional shape.

It should be noted that this approach remains a heuristic. The choice of moment indices is not optimised in any principled way, and the gap between $\mathcal{K}_1$ and $\mathcal{K}_2$ shows that the result is sensitive to the specific selection.

5.6 Wasserstein Distance of the Moment-of-Moments

A different approach is explored here as an alternative to the SDP method. The idea is to apply the moment computation a second time: the first 50 raw moments of the sample set $\{x_1, \ldots, x_n\}$ at each pixel are computed to produce a vector $\mathbf{m} = (m_1, m_2, \ldots, m_{50})$. This vector is then treated as a new set of 50 numerical values, and its own first 50 moments are

computed. The Wasserstein distance is finally applied to these second-level moment vectors at each pixel, using the same empirical quantile formula as in Section ??.

Figure 7 shows the result. The biased region stands out very clearly in all zones where the sample count exceeds 1 000 per pixel, making this the method that best identifies the biased columns among all the approximation strategies tested. In the outer border where pixels have only around 100 samples, the signal remains weak, as was also observed with the SDP approach. The sampling heterogeneity between the concentric zones is only partially visible.

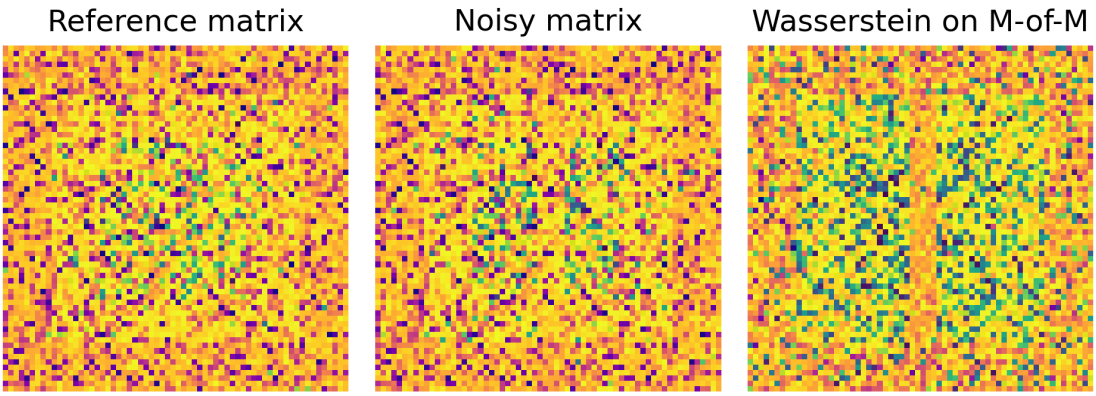

Figure 4: Pixelwise map produced by the Wasserstein distance computed from the moments of the moment vector.

Compared to the SDP approach of Section 6.4, this method has two practical advantages. First, it is approximately 100 times faster in our experiments, since it requires only moment computations and a sorting step, with no optimisation solver involved. Second, the biased region appears more clearly in the resulting map, which makes it easier to interpret in practice.

This result is an empirical observation. No theoretical framework currently explains why this construction recovers the distributional signal so effectively, and no existing study in the optimal transport or dosimetry literature appears to have investigated this approach. The result is reproducible across different random seeds and bias levels in our experiments, which suggests that it is not a numerical artefact. Understanding why the moment-of-moments construction works would be a natural direction for future work.

5.7 Summary of Results

Table 2 summarises the ability of each method to detect the two types of discrepancy present in the simulated dose maps: the additive bias (columns 30 to 34) and the sampling heterogeneity (border region with low sample counts).

Method	**Bias detection**	**Sampling heterogeneity**
MAE / MSE	Partial	No
SSIM	No	No
Gamma index	No	No
T-test	No	No
Wasserstein (raw samples)	Yes	Yes
Wasserstein (moment-SoS, $r = 2$)	Partial	Partial
MAE of moment subsets	Partial	Partial
Wasserstein (moment-of-moments)	Yes	Partial

Table 2: Summary of the ability of each method to detect the additive bias and the sampling heterogeneity in the simulated dose maps. The Wasserstein distance computed from raw samples serves as the reference target.

The main finding is that classical metrics such as MAE and MSE show only a weak and unreliable signal in the biased region, while the other classical metrics provide no useful information at all. The Wasserstein distance computed from raw samples clearly outperforms all of them, detecting both the biased region and the sampling heterogeneity in a robust way. Among the approximation strategies based on compressed representations, both the moment-SoS approach and the moment-of-moments approach successfully identify the biased region. The moment-of-moments method additionally recovers the sampling heterogeneity with a clarity comparable to the raw-sample reference, while being significantly faster than the SDP-based approach. These results suggest that moment-based approximations are a viable path toward making the Wasserstein distance practical for clinical dosimetry.

6 Conclusion

7 Bibliography

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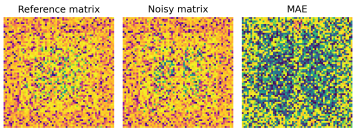
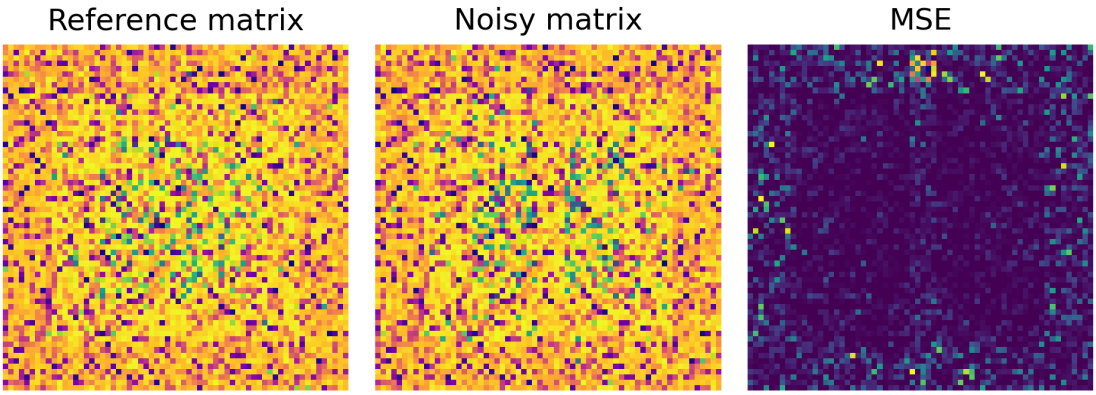
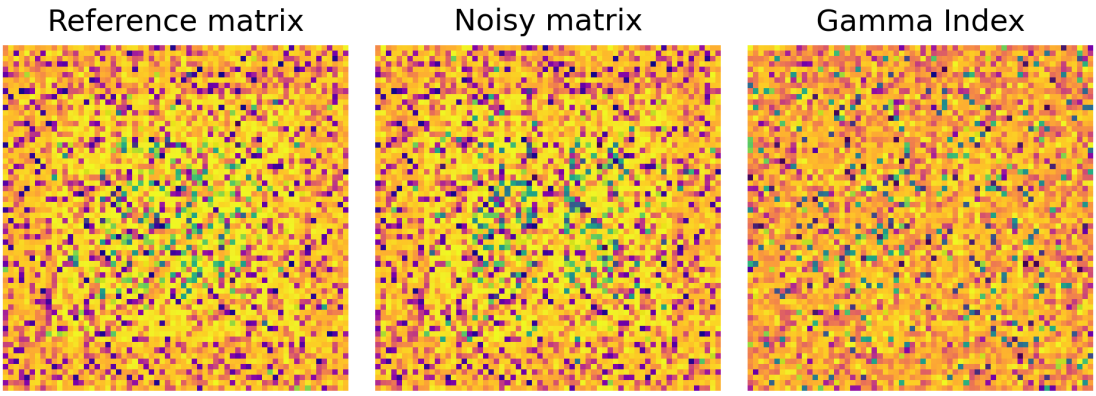
[4] O. Mula and A. Nouy, *Moment-SOS approximation of the optimal transport problem*, preprint, 2024.

8 Annexe


Reference matrix
Noisy matrix
MAE


Reference matrix
Noisy matrix
MSE


Reference matrix
Noisy matrix
Gamma Index

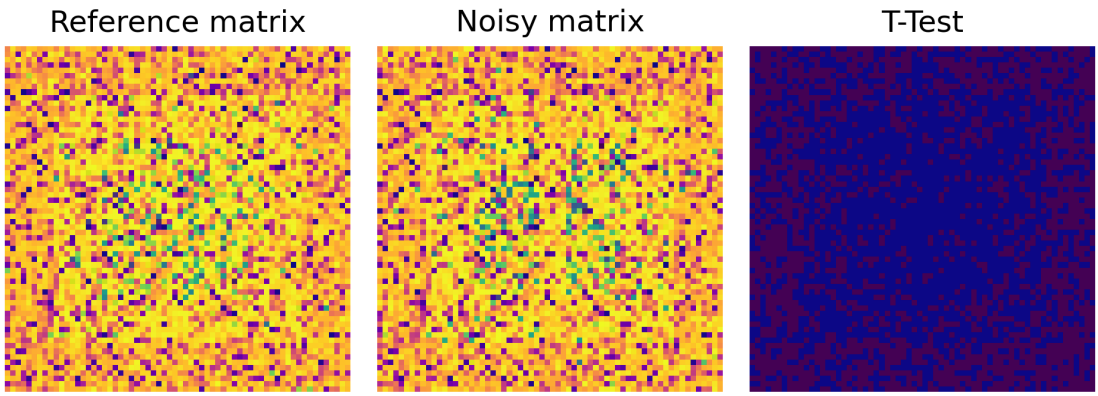


Figure 5: Pixelwise maps produced by the classical metrics (MAE, MSE, SSIM, gamma index, t-test) for the weak bias setting ($b = 0.05$).

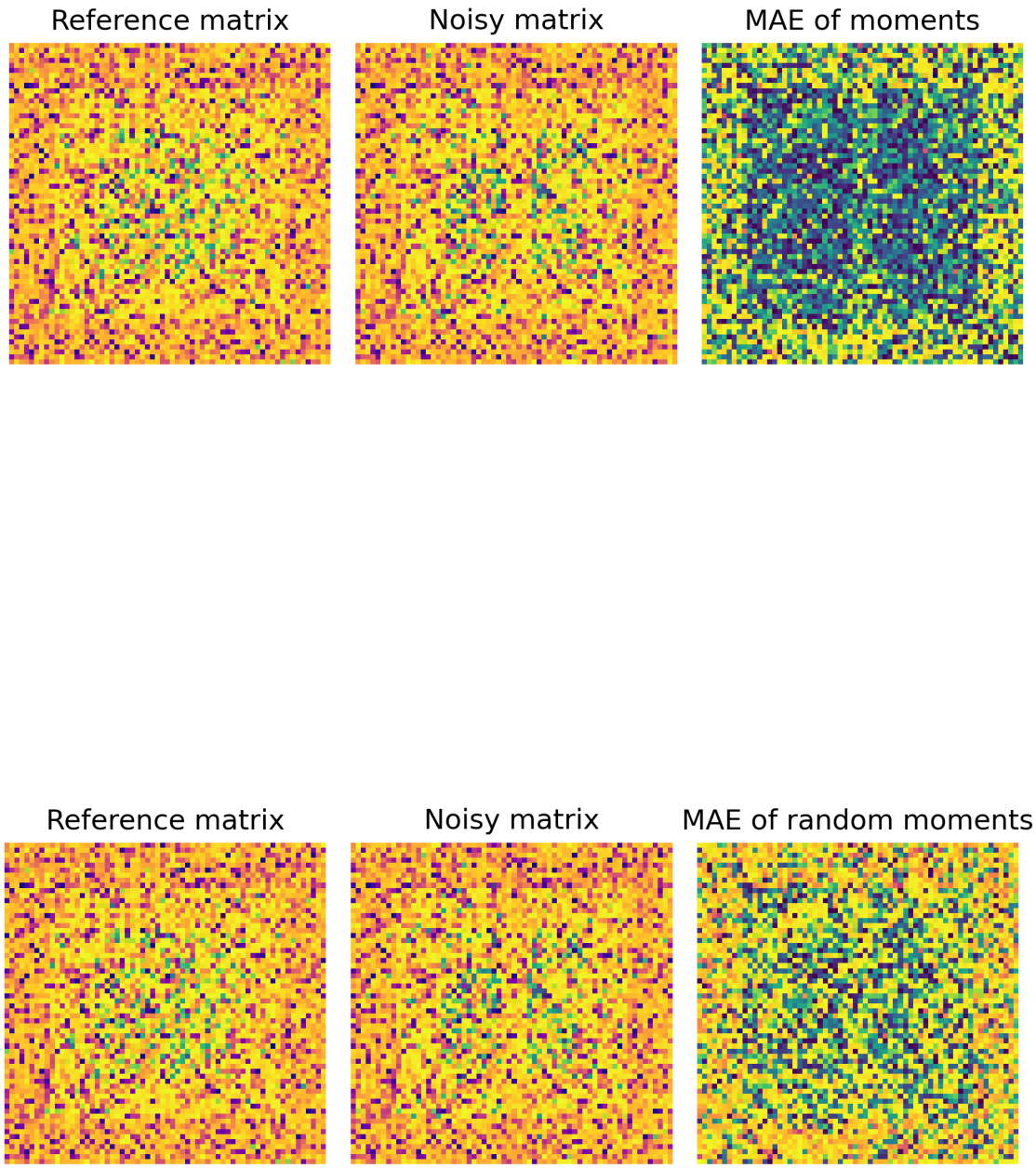


Figure 6: Comparison maps produced by the MAE of averaged moment subsets, for evenly spaced indices $\mathcal{K}_1$ (above) and randomly chosen indices $\mathcal{K}_2$ (below).